# Hydrohalogenation Reactions of 1,2-Allenic Sulfones. A Novel Synthesis of 2-Haloallylic Sulfones

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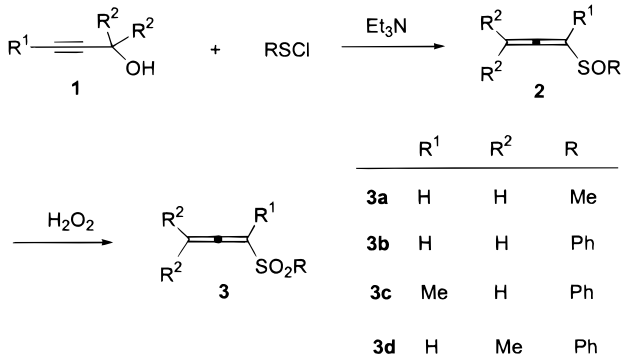
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## Introduction

Hydrohalogenation reactions of electron-deficient C−C triple bonds, i.e., 2-alkynoic acids,[1] 2-alkynonates,[1] 2-alkynamides,[1] 2-propyne nitriles,[1] 2-alkynones,[1] 1-alkynyl phosphonates,[2] 1-alkynyl sulfoxides,[3] 1-alkynyl triflone,[4] 1-alkynyl iodinium salts,[5] etc., have become a very powerful tool for the highly stereoselective synthesis of 3-halo-(2*Z*)-alkenoic acids and the corresponding equivalents. Another class of compounds attracting our interest is electron-deficient 1,2-dienes. Due to the unique arrangement of the two sets of $\pi$-orbitals, the reactivity of electron-deficient 1,2-dienes should be higher than that of the corresponding alkynes.[6] Recently, we developed a methodology for the efficient syntheses of 3-halo-3-butenoic acids/methyl esters and 4-halo-4-penten-2-ones via the hydrohalogenation reactions of 2,3-butadienoic acid/methyl ester and 3,4-pentadien-2-one, respectively.[7] In this paper, we wish to disclose our recent results on the corresponding hydrohalogenation reactions of 1,2-allenic sulfones.

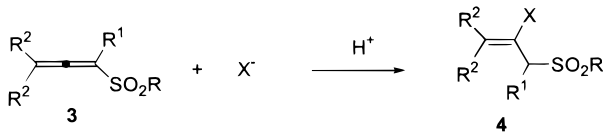
The known methods for the synthesis of these compounds are (1) the oxidation of 2-chloroallyl sulfide, which could be prepared via the reaction of thiophenols with the corresponding 2,3-dihalopropenes;[8] (2) the reaction of 1-(phenylsulfonyl)propyne with hydrazonyl chlorides to afford a mixture of the allylic sulfone, the vinylic sulfone, and many other products;[9] (3) the radical addition of alkylsulfonyl iodide with allenes to produce a mixture of several adducts;[10] (4) the reaction of 2-haloallyl iodides with $PhSO_2Na$.[11] It should be noted that 2-haloallyl iodides, especially the bromo and iodo analogues, are not stable and usually are not readily available. Thus, a general, clean, and high-yielding method for the efficient synthesis of the title compounds is highly desirable.

**Scheme 1**




| | $R^1$ | $R^2$ | R |
|---|---|---|---|
| **3a** | H | H | Me |
| **3b** | H | H | Ph |
| **3c** | Me | H | Ph |
| **3d** | H | Me | Ph |

**Scheme 2**[a]




X = Cl, Br, I

[a] Conditions A: MX (M = Na or Li), HOAc (solvent). Conditions B: MX (M = Na or Li), HOAc (1 equiv), $CH_3CN$ (solvent). Conditions C: HX (as the solvent and the reagent).

## Results and Discussion

1,2-Allenic sulfones are readily available from the oxidation of their corresponding sulfoxides, which can be easily prepared from the reaction of propargylic alcohols with RSCl (Scheme 1).[12] Four such compounds were synthesized via the chemistry shown in Scheme 1.

Three sets of reaction conditions were established to carry out the hydrohalogenation reactions of 1,2-allenic sulfones with metallic halides (Scheme 2). Some typical results are summarized in Table 1 and Table 2.

From the results of the hydrohalogenation reactions shown in Tables 1 and 2, as well as the results with allenic ketones, carboxylic acids, and esters, the following conclusions are obvious. (1) The electron-withdrawing ability of $SO_2R$ is somewhat similar to that of the carbonyl group in esters and is much weaker than that of keto groups (compare the results with that in ref 7).[13] A very slow reaction of **3a** with LiI in HOAc at room temperature was observed. (2) The migration of the C=C bond in product **4ac** was not observed even when **4ac** was heated in HOAc at 118 °C for 11 h.[7,9] (3) For the synthesis of the iodides, both NaI and LiI worked well, although the yield with LiI was higher under the same reaction conditions (compare entries 1 with 2 in Table 1), while for bromides and chlorides, lithium salts afforded the products in much higher yields than those

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**Table 1. Hydrohalogenation Reactions of 1,2-Allenic Sulfone (3a)**

| entry | MX (equiv) | condns | temp (°C) | time (h) | product | yield (%)[a] |
|---|---|---|---|---|---|---|
| 1 | NaI (1.1−2) | A | 80 | 0.5 | **4aa** | 80 |
| 2 | LiI (1.7) | A | reflux | 0.5 | **4aa** | 87 |
| 3 | NaBr (2) | A | reflux | 4 | **4ab** | 61[b] |
| 4 | $LiBr·H_2O$ (2) | A | reflux | 1.5 | **4ab** | 79 |
| 5 | NaCl (4.4) | A | reflux | 11 | **4ac** | 28[c] |
| 6 | $LiCl·H_2O$ (5) | A | reflux | 11 | **4ac** | 83 |
| 7 | LiI (2) | B | 80 | 12 | **4aa** | 76 |
| 8 | $LiBr·H_2O$ (2) | B | 80 | 12 | **4ab** | 76 |
| 9 | $LiCl·H_2O$ (2) | B | 80 | 12 | **4ac** | 85 |
| 10 | HI[d] | C | 80 | 8 | **4aa** | 65 |
| 11 | HCl[e] | C | 80 | 8 | **4ac** | 83 |

[a] Isolated yield, unless otherwise stated. [b] 30% of **3a** was recovered. [c] 70% of **3a** was recovered. [d] Aqueous acid (45%) was used. [e] Aqueous acid (37%) was used.

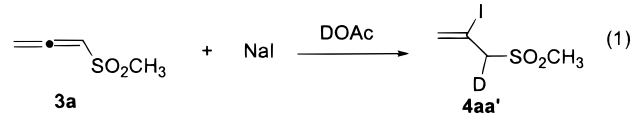
**Table 2. Hydrohalogenation Reactions of 3b−d**

| entry | 3 | MX (equiv) | condns | temp (°C) | time (h) | product | yield (%)[a] |
|---|---|---|---|---|---|---|---|
| 1 | **3b** | NaI (2) | A | 80 | 0.5 | **4ba** | 82 |
| 2 | **3b** | $LiBr·H_2O$ (2) | A | 90 | 0.5 | **4bb** | 78 |
| 3 | **3b** | NaBr (5) | A | 90 | 8 | **4bb** | 67 |
| 4 | **3b** | $LiCl·H_2O$ (2) | A | 90 | 2 | **4bc** | 69 |
| 5 | **3b** | NaCl (5) | A | 90 | 13 | **4bc** | 14[b] |
| 6 | **3b** | LiI (2.5) | B | 80 | 12 | **4ba** | 76 |
| 7 | **3b** | $LiBr·H_2O$ (2.5) | B | 80 | 12 | **4bb** | 67 |
| 8 | **3b** | $LiCl·H_2O$ (2.5) | B | 80 | 12 | **4bc** | 76 |
| 9 | **3b** | HI[c] | C | 80 | 0.5 | **4ba** | 44 |
| 10 | **3b** | HBr[d] | C | 80 | 22 | **4bb** | 53 |
| 11 | **3b** | HCl[e] | C | 80 | 22 | **4bc** | 81 |
| 12 | **3c** | NaI (3.5) | A | 80 | 4 | **4ca** | 88 |
| 13 | **3c** | NaI (1.2) | A | reflux | 4 | **4ca** | 83 |
| 14 | **3c** | $LiBr·H_2O$ (5) | A | 110 | 20 | **4cb** | 77 |
| 15 | **3c** | $LiBr·H_2O$ (2) | A | reflux | 10 | **4cb** | 74 |
| 16 | **3c** | NaBr (5) | A | 110 | 20 | **4cb** | 41[f] |
| 17 | **3d** | NaI (3) | A | reflux | 2 | **4da** | 74 |
| 18 | **3d** | NaI (1.2) | A | reflux | 2.5 | **4da** | 69 |

[a] Isolated yield, unless otherwise stated. [b] 36% of **3b** was recovered. [c] Aqueous acid (45%) was used. [d] Aqueous acid (40%) was used. [e] Aqueous acid (37%) was used. [f] NMR yield.

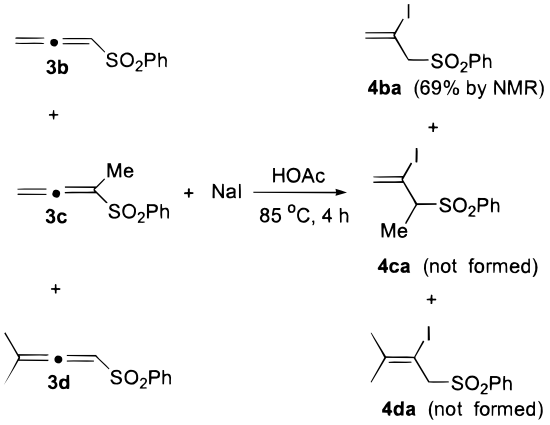
of the sodium salts (compare entries 3 with 4 and 5 with 6 in Table 1, compare entries 2 with 3, 4 with 5, and 14 with 16 in Table 2). (4) The reaction can proceed in $CH_3CN$ using only 1 equiv of HOAc (conditions B). (5) The reaction of HX with 1,2-allenic sulfones also afforded the hydrohalogenation products in reasonable yields (entries 10 and 11 in Table 1, entries 9−11 in Table 2).

This reaction proceeded via a nucleophilic addition of $X^-$ to the $\alpha,\beta$ C=C bond of the two C=C bonds in 1,2-allenic sulfones.[7] The proton came from HOAc or HX; thus, methyl 1-deuterio-2-iodo-2-propenyl sulfone (**4aa′**) could be conveniently prepared via the reaction of **3a** with NaI in DOAc in 77% yield. The extent of D-incorporation is >97% as determined by 300 MHz $^1H$ NMR and 75.48 MHz $^{13}C$ NMR spectra (eq 1).




The methyl group at the α-position greatly slowed the hydrohalogenation reaction, probably due to the increase of steric hindrance at the α-position to pick up a proton from either HOAc or HX (compare Tables 1 and 2). Even the two methyl groups in **3d** affected the rate of this reaction. A reaction of a mixture of **3b** (0.1 mmol), **3c** (0.1 mmol), and **3d** (0.1 mmol) with NaI (0.1 mmol) in HOAc at 85 °C for 4 h afforded the hydroiodination product **4ba** in 69% NMR yield, while the formations of **4ca** and **4da** were not observed during the specified reaction time (Scheme 3).

**Scheme 3**




In summary, the hydrohalogenation reaction of 1,2-allenic sulfones afforded 2-haloallyl sulfones in high yields using conditions A, B, and C. Allylic sulfones are a class of compounds with significant synthetic potential: (1) the α-positions can be regioselectively alkylated;[14] (2) the sulfone groups can be easily removed by reduction;[15] (3) the nucleophilic ($S_N2'$) and radical ($S_H2'$) allylic substitution reactions can take place.[16] Vinyl halides, especially iodides and bromides, are novel starting materials for radical reactions and transition metal catalyzed C−C bond formation reactions. All these make 2-haloallylic sulfones prepared by the method described in this paper valuable synthetic intermediates. The investigation regarding the synthetic utility of these compounds is currently being carried out in our laboratory.

## Experimental Section

**General.** NaI, LiI, NaBr, $LiBr·H_2O$, NaCl, $LiCl·H_2O$, hydrochloric acid, hydrobromic acid, hydriodic acid, and acetic acid are commercially available and used as is. $^1H$ NMR spectra were measured using $CDCl_3$ as the solvent and $Me_4Si$ as the internal standard. The NMR yields were measured using $CH_2Br_2$ as the internal standard. 1-(Methylsulfonyl)-1,2-propadiene (**3a**), (1,2-propadienylsulfonyl)benzene (**3b**), (1-methyl-1,2-propadienylsulfonyl)benzene (**3c**), and (3-methyl-1,2-butadienylsulfonyl)benzene (**3d**) were prepared according to the published procedure.[12,17]

**Hydrohalogenation Reactions of 1,2-Allenic Sulfones. (a) Synthesis of 2-Iodo-3-(methylsulfonyl)-1-propene (4aa). Typical Procedure. Conditions A**. A solution of 1-(methylsulfonyl)-1,2-propadiene (**3a**) (118 mg, 1 mmol) and NaI (165 mg, 1.1 mmol) in HOAc (0.5 mL) was stirred at 80 °C for 1.5 h. After complete conversion of the starting material as monitored by TLC (eluent: petroleum ether/ethyl acetate = 1:1), the mixture was quenched with water (3 mL), neutralized with $NaHCO_3$, and then extracted with ether (4 × 10 mL). The combined ether layer was dried over $MgSO_4$. Evaporation of the

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solvent and flash chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 2:1) afforded 197 mg (80%) of **4aa**: solid, mp 56−57 °C (dichloromethane/petroleum ether) (reported: [10] mp 57−58 °C); $^1$H NMR 6.57 (d, $J$ = 1.5 Hz, 1H), 6.25 (d, $J$ = 1.5 Hz, 1H), 4.12 (s, 2H), 3.00 (s, 3H); MS *m/e* 247 ($M^+$ + 1, 12.12), 246 ($M^+$, 4.15), 167 (100); IR (KBr) 1600, 1294, 1118 cm$^{-1}$.

**Typical Procedure. Conditions B.** A solution of 1-(methylsulfonyl)-1,2-propadiene (**3a**) (59 mg, 0.5 mmol), HOAc (0.03 mL, 0.52 mmol), and LiI (134 mg, 1 mmol) in acetonitrile (1 mL) was stirred at 80 °C for 12 h. After complete conversion of the starting material, the mixture was quenched with water (1 mL), neutralized with $NaHCO_3$, and then extracted with ether (4 × 5 mL). The combined ether layer was dried over $MgSO_4$. Evaporation of the solvent and flash chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 2:1) afforded 93 mg (76%) of **4aa**.

**Typical Procedure. Conditions C.** A solution of 1-(methylsulfonyl)-1,2-propadiene (**3a**) (59 mg, 0.5 mmol) and hydriodic acid (45%, 1 mL) was stirred at 80 °C for 8 h. After complete conversion of the starting material, the mixture was quenched with water (1 mL), neutralized with $NaHCO_3$, and then extracted with ether (4 × 5 mL). The combined ether layer was dried over $MgSO_4$. Evaporation of the solvent and flash chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 2:1) afforded 80 mg (65%) of **4aa**.

The following compounds were prepared similarly. The details of reactions under conditions A are listed below.

**(b) 2-Bromo-3-(methylsulfonyl)-1-propene (4ab).** Using **3a** (60 mg, 0.51 mmol) and LiBr·$H_2O$ (104 mg, 1 mmol) afforded 80 mg (79%) of **4ab**:[18] liquid; $^1$H NMR $\delta$ 6.12 (d, $J$ = 2 Hz, 1H), 5.94 (d, $J$ = 2 Hz, 1H), 4.09 (s, 2H), 3.04 (s, 3H); MS *m/e* 201 ($M^+$ + 1 ($^{81}$Br), 31.42), 199 ($M^+$ + 1 ($^{79}$Br), 31.55), 119 (100); IR (neat) 1622, 1304, 1128 cm$^{-1}$.

**(c) 2-Chloro-3-(methylsulfonyl)-1-propene (4ac).** Using **3a** (25 mg, 0.21 mmol) and LiCl·$H_2O$ (60 mg, 1 mmol) afforded 27.2 mg (83%) of **4ac**: liquid; $^1$H NMR $\delta$ 5.66 (d, $J$ = 2 Hz, 1H), 5.65 (d, $J$ = 2 Hz, 1H), 3.96 (s, 2H), 3.02 (s, 3H); MS *m/e* 157 ($M^+$ + 1($^{37}$Cl), 2.42), 156 ($M^+$ ($^{37}$Cl), 1.61), 155 ($M^+$ + 1($^{35}$Cl), 6.71), 154 ($M^+$ ($^{35}$Cl), 3.50), 75 (100); IR (neat) 1630, 1306, 1134 cm$^{-1}$; HRMS calcd for $C_4H_7O_2{}^{37}ClS$ 155.9826, found 155.9871; calcd for $C_4H_7O_2{}^{35}ClS$ 153.9855, found 153.9827.

**(d) ((2-Iodo-2-propenyl)sulfonyl)benzene (4ba).** Using **3b** (36 mg, 0.2 mmol) and NaI (60 mg, 0.4 mmol) afforded 51 mg (82%) of **4ba**: solid, mp 90.5−91.5 °C (dichloromethane/petroleum ether) (reported:[10] mp 89.5−92 °C); $^1$H NMR $\delta$ 7.93 (d, $J$ = 7 Hz, 2H), 7.68 (t, $J$ = 7 Hz, 1H), 7.58 (t, $J$ = 7 Hz, 2H), 6.27 (q, $J$ = 2 Hz, 1H), 6.08 (d, $J$ = 2 Hz, 1H), 4.16 (d, $J$ = 0.9 Hz, 2H); MS *m/e* 308 ($M^+$, 7.40), 77 (100); IR (KBr) 1612, 1306, 1134 cm$^{-1}$.

**(e) ((2-Bromo-2-propenyl)sulfonyl)benzene (4bb).** Using **3b** (36 mg, 0.2 mmol) and LiBr·$H_2O$ (42 mg, 0.4 mmol) afforded 41 mg (78%) of **4bb**: solid, mp 75.5−76.5 °C (dichloromethane/petroleum ether); $^1$H NMR $\delta$ 7.93 (d, $J$ = 6 Hz, 2H), 7.68 (t, $J$ = 6 Hz, 1H), 7.57 (t, $J$ = 6 Hz, 2H), 5.82 (d, $J$ = 2 Hz, 1H), 5.77 (d, $J$ = 2 Hz,1H), 4.13 (s, 2H); MS *m/e* 263 ($M^+$ + 1 ($^{81}$Br), 1.69), 261 ($M^+$ + 1 ($^{79}$Br), 1.68), 77 (100); IR (KBr) 1624, 1306, 1134 cm$^{-1}$; HRMS calcd for $C_9H_9O_2{}^{79}BrS$ 259.9507, found 259.9542.

**(f) ((2-Chloro-2-propenyl)sulfonyl)benzene (4bc).** Using **3b** (36 mg, 0.2 mmol) and LiCl·$H_2O$ (24 mg, 0.4 mmol) afforded 30 mg (69%) of **4bc**: solid, mp 54−55.5 °C (dichloromethane/petroleum ether) (reported:[8] mp 57−58 °C); $^1$H NMR $\delta$ 7.88 (d, $J$ = 7 Hz, 2H), 7.63 (t, $J$ = 7 Hz, 1H), 7.52 (t, $J$ = 7 Hz, 2H), 5.45 (s, 1H), 5.32 (s, 1H), 4.03 (s, 2H); MS *m/e* 219 ($M^+$ + 1 ($^{37}$Cl), 1.07), 217 ($M^+$ + 1 ($^{35}$Cl), 2.80), 77 (100); IR (KBr) 1628, 1320, 1150 cm$^{-1}$.

**(g) ((2-Iodo-1-methyl-2-propenyl)sulfonyl)benzene (4ca).** Using **3c** (29 mg, 0.15 mmol) and NaI (80 mg, 0.53 mmol) afforded 42 mg (88%) of **4ca**: liquid; $^1$H NMR $\delta$ 7.89 (d, $J$ = 7 Hz, 2H), 7.67 (t, $J$ = 7 Hz, 1H), 7.55 (t, $J$ = 7 Hz, 2H), 6.31 (t, $J$ = 2 Hz, 1H), 6.07 (d, $J$ = 2 Hz, 1H), 3.70(q, $J$ = 7 Hz, 1H), 1.55 (d, $J$ = 7 Hz, 3H); MS *m/e* 322 ($M^+$, 8.00) 53 (100); IR (neat) 1600, 1300, 1140 cm$^{-1}$; HRMS calcd for $C_{10}H_{11}O_2IS$ 321.9523, found 321.9532.

**(h) ((2-Bromo-1-methyl-2-propenyl)sulfonyl)benzene (4cb).** Using **3c** (21 mg, 0.11 mmol) and LiBr·$H_2O$ (52 mg, 0.5 mmol) afforded 23 mg (77%) of **4cb**: liquid; $^1$H NMR $\delta$ 7.4−7.9 (m, 5H), 5.71 (d, $J$ = 2 Hz, 1H), 5.63 (d, $J$ = 2 Hz, 1H), 3.85 (q, $J$ = 7 Hz, 1H), 1.50 (d, $J$ = 7 Hz, 3H); MS *m/e* 276 ($M^+$ ($^{81}$Br), 63.02), 274 ($M^+$ ($^{79}$Br), 60.38), 125 (100); IR (neat) 1608, 1300, 1140 cm$^{-1}$; HRMS calcd for $C_{10}H_{11}O_2{}^{81}BrS$ 275.9643, found 275.9595; calcd for $C_{10}H_{11}O_2{}^{79}BrS$ 273.9663, found 273.9662.

**(i) ((2-Iodo-3-methyl-2-butenyl)sulfonyl)benzene (4da).** Using **3d** (36 mg, 0.17 mmol) and NaI (80 mg, 0.53 mmol) afforded 43 mg (74%) of **4da**: solid; mp 85−86 °C (dichloromethane/petroleum ether); $^1$H NMR $\delta$ 7.87 (d, $J$ = 7 Hz, 2H), 7.65 (t, $J$ = 7 Hz, 1H), 7.54 (t, $J$ = 7 Hz, 2H), 4.37 (s, 2H), 1.91 (s, 3H), 1.65 (s, 3H); MS *m/s* 336 ($M^+$, 12.74), 209 (100); IR (neat) 1616, 1308, 1136 cm$^{-1}$; HRMS calcd for $C_{11}H_{13}O_2IS$ 335.9680, found 335.9687.

**Reaction of 3a with NaI in DOAc. Synthesis of 3-Deuterio-2-iodo-3-(methylsulfonyl)-1-propene (4aa′).** Using **3a** (118 mg, 1 mmol), NaI (300 mg, 2 mmol), and DOAc (1 mL) afforded 191 mg (77%) of **4aa′**: liquid; $^1$H NMR $\delta$ 6.56 (d, $J$ = 1 Hz, 1H), 6.24 (d, $J$ = 1 Hz, 1H), 4.10 (s, 1H), 2.99 (s, 3H); $^{13}$C NMR $\delta$ 136.381, 88.412, 68.009 (t, $J_{D-C}$ = 21.44 Hz), 41.071.

**Comparison of Reactivities of 3b, 3c, and 3d.** A solution of **3b** (11.8 mg, 0.1 mmol), **3c** (19.4 mg, 0.1 mmol), **3d** (20.8 mg, 0.1 mmol), and NaI (15 mg, 0.1 mmol) in HOAc (1.5 mL) was stirred at 85 °C for 4 h. After usual workup, the $^1$H NMR spectrum of the crude product showed **4ba** was formed in 69% yield by NMR. The formations of **4ca** and **4da** were not observed.

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**Supporting Information Available:** $^1$H NMR spectra of the compounds **4ac**, **4bb**, **4ca**, **4cb**, **4da**, and **4aa′** as well as $^{13}$C NMR spectrum of the compound **4aa′** (8 pages). See any current masthead page for ordering and Internet access information.

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